

# EC614: Macroeconomic Forecasting

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## Section A: Multiple Choice Questions (25 marks)

Select the best answer for each question.

### Question 1

In a reduced-form VAR, the error terms across equations are:

- a) Always uncorrelated by construction
- b) May be correlated if variables are related
- c) Always normally distributed
- d) Independent of past values

### Question 2

The primary advantage of ARDL over traditional cointegration methods is:

- a) It requires all variables to be  $I(1)$
- b) It works with variables of mixed integration orders  $I(0)$  and  $I(1)$
- c) It is computationally simpler
- d) It doesn't require lag selection

### Question 3

In Granger causality testing, if X Granger-causes Y, it means:

- a) X definitely causes Y in a causal sense
- b) Past values of X help predict Y beyond Y's own past
- c) Y cannot Granger-cause X simultaneously
- d) X and Y must be cointegrated

### Question 4

The Pesaran-Shin-Smith (2001) bounds test in ARDL tests the null hypothesis:

- a) All variables are  $I(1)$
- b) Variables are cointegrated
- c) No long-run relationship exists
- d) Variables are stationary

### Question 5

In a structural VAR (SVAR), identification restrictions are imposed on:

- a) The lag length only
- b) The contemporaneous relationships among variables
- c) The forecast horizon
- d) The variance of residuals

### Question 6

Which statement about impulse response functions (IRFs) is FALSE?

- a) They trace dynamic effects of shocks over time
- b) They require the VAR to have uncorrelated errors
- c) Confidence bands indicate statistical uncertainty
- d) They are always symmetric between variables

### Question 7

In a recursive VAR, the ordering of variables matters because:

- a) It determines computational efficiency
- b) It imposes assumptions about contemporaneous effects
- c) It affects lag length selection
- d) It changes the sample size

### Question 8

The forecast error variance decomposition (FEVD) answers:

- a) How accurate the forecast is
- b) What fraction of forecast error variance is due to each shock
- c) Whether the model is correctly specified
- d) If variables are cointegrated

### Question 9

If two  $I(1)$  variables are cointegrated, their linear combination is:

- a)  $I(1)$
- b)  $I(2)$
- c)  $I(0)$
- d) Non-stationary with no defined order

### Question 10

The error correction term in a VECM represents:

- a) The forecasting error
- b) Deviation from long-run equilibrium
- c) Short-run dynamics
- d) Measurement error

### Question 11

In the Johansen cointegration test, the trace statistic tests:

- a) Whether  $\text{rank}(\Pi) = 0$
- b) Whether  $\text{rank}(\Pi) \leq r$  against  $\text{rank}(\Pi) > r$
- c) Stationarity of individual series
- d) Granger causality

**Question 12**

Weak exogeneity in a VECM context means:

- a) The variable doesn't respond to disequilibrium
- b) The variable is determined outside the model
- c) The variable is stationary
- d) The variable has weak predictive power

**Question 13**

The primary problem with estimating a VAR with non-stationary I(1) variables that are NOT cointegrated is:

- a) Computational complexity
- b) Spurious regression
- c) Loss of degrees of freedom
- d) Heteroskedasticity

**Question 14**

In ARDL modeling, if the F-statistic falls between the lower and upper bounds:

- a) We reject cointegration
- b) We accept cointegration
- c) The result is inconclusive
- d) We need to increase lag length

**Question 15**

The Cholesky decomposition in recursive VAR estimation:

- a) Tests for cointegration
- b) Orthogonalizes the VAR residuals
- c) Selects optimal lag length
- d) Estimates structural parameters

**Question 16**

Compared to univariate ARIMA models, VARs:

- a) Are always superior for forecasting
- b) Capture cross-variable dynamics
- c) Require less data
- d) Have clearer structural interpretation

**Question 17**

In panel data with fixed effects, the "within" transformation:

- a) Adds time-specific effects
- b) Removes individual-specific means
- c) Differences all variables
- d) Standardizes variables

**Question 18**

The Hausman test in panel data compares:

- a) Fixed effects vs pooled OLS
- b) Random effects vs pooled OLS
- c) Fixed effects vs random effects
- d) Static vs dynamic models

**Question 19**

In SVAR identification using the Blanchard-Perotti approach:

- a) Only short-run restrictions are used
- b) Institutional knowledge provides parameter values
- c) All variables must be ordered recursively
- d) Long-run restrictions are necessary

**Question 20**

When VAR residuals show autocorrelation, the most likely issue is:

- a) Too many lags included
- b) Insufficient lag length
- c) Non-stationarity
- d) Heteroskedasticity

## Part B: Short Answer Questions (30 marks)

Answer all questions. Write concisely but completely.

**Question 1 (5 marks)**

Explain the difference between reduced-form VAR, recursive VAR, and structural VAR. Under what circumstances would you prefer using each type? Provide a concrete example from macroeconomic policy analysis.

**Question 2 (5 marks)**

You estimate an ARDL(2,4,0,0,0,4,4,1) model for a relationship between multiple variables.

- (a) What do the numbers in parentheses represent?
- (b) The bounds test F-statistic = 6.45 (critical values:  $I(0) = 2.45$ ,  $I(1) = 3.61$  at 5% level). What do you conclude about cointegration?
- (c) If cointegration exists, what is your next step in the analysis?

**Question 3 (5 marks)**

Consider a three-variable VECM for money (M), prices (P), and GDP (Y) for Pakistan.

The Johansen test indicates  $r = 1$  cointegrating relationship. The estimated cointegrating vector is:

$$\beta' = [1, -0.85, -0.35] \text{ for } [M, P, Y]$$

- (a) Interpret this long-run relationship economically.
- (b) What does  $r = 1$  tell you about the system?
- (c) How would you test if money (M) is weakly exogenous?

**Question 4 (15 marks)**

You are analyzing monetary policy transmission in Pakistan using quarterly data from 2000Q1 to 2023Q4 for the following variables:

- GDP growth rate (Y)
- CPI inflation ( $\pi$ )
- Policy interest rate (R)

- Real effective exchange rate (REER)

**Part (a): Preliminary Analysis (5 marks)**

(i) Explain the steps you would take to prepare the data before estimating a VAR model.

(ii) What unit root tests would you apply and why?

(iii) How would you handle the different orders of integration if some variables are  $I(0)$  and others are  $I(1)$ ?

**Part (b): VAR Specification (5 marks)**

(i) Explain how you would select the optimal lag length for the VAR.

(ii) Would you include deterministic terms (constant, trend)? Justify your answer.

(iii) Write out the VAR equations in matrix form for lag order  $p=2$ .

**Part (c): Structural Identification (5 marks)**

(i) Propose a recursive ordering for these variables based on institutional knowledge of Pakistan's economy. Justify your ordering.

(ii) Alternatively, propose one short-run restriction and one long-run restriction for SVAR identification.

(iii) What economic shock are you trying to identify, and why does it matter for policy?